Question	Answer	Marks	Guidance
1(a)	acceleration is (always) perpendicular to velocity	**B1**	
	velocity and acceleration have constant magnitude **or** velocity and acceleration change direction	**B1**	Accept in terms of either velocity or acceleration (both not needed). 'Constant speed', with no reference to magnitude of velocity, is insufficient. **Allow** 'acceleration is (always) directed towards the centre of the circle'. **Allow** 'velocity is (always) tangential to the circle'.
1(b)(i)	only other force is weight, which is vertical **or** weight has no horizontal component **or** vertical component of lift is equal (and opposite) to the weight	**B1**	**Allow** 'gravitational force' for 'weight'. **Allow** 'balances' or 'cancels' for 'equal to'.
	lift must have a horizontal component to cause centripetal acceleration	**B1**	**Allow** 'Horizontal component of lift provides / is the resultant / centripetal force'.
1(b)(ii)	arrow drawn from aircraft, horizontally to the left	**B1**	Arrow does not need to touch the aircraft but be pointing away from the aircraft and pass through some part of the aircraft when extended. **Allow** tolerance ± 5°. Treat any other arrows that (extended if necessary) pass through some part of the aircraft as contradictory unless labelling clearly conveys that one is the resultant force or the other arrows are not.
1(b)(iii)	$a = v^2 / r$	**C1**	**Allow** $a = \omega^2 r$ **and** $v = r\omega$.
	$a = 180^2 / 7200$ $= 4.5\ \text{m s}^{-2}$	**A1**	Correct to at least 2 significant figures. AFC.

Question	Answer	Marks	Guidance
1(b)(iv)	vertical component of lift balances the weight **or** horizontal component of lift causes the centripetal acceleration **or** $mg = L\cos\theta$ **and** $ma = L\sin\theta$	B1	**Allow** 'equals' for 'balances'. **Allow** 'provides … force' for 'causes … acceleration'. **Allow** F for L. **Allow** 'W' for 'mg' and 'F_C' or 'mv^2/r' for 'ma'.
	$\tan\theta = a/g$	C1	
	$\tan\theta = 4.5/9.81$ $\theta = 25°$	A1	Possible ECF from **1(b)(iii)**. Correct to at least 2 significant figures. (24.6)